

STUDENT PERFORMANCE ANALYTICS & DECISION SUPPORT SYSTEM

1. Purpose:

Business rules define the academic logic and conditions used by the analytics system.

2. Student Management Rules

Rule ID	Business Rule
BR001	Every student must have a unique student ID
BR002	Student details must be validated before storage
BR003	Student records cannot be duplicated

3. Assessment Rules

Rule ID	Business Rule
BR004	Every test must have a unique Test ID
BR005	Each question must belong to one subject and chapter
BR006	Question difficulty must be categorized as Easy, Medium, or Hard

4. Score Calculation Rules

Rule ID	Business Rule
BR007	$\text{Percentage} = (\text{Correct Answers} / \text{Total Questions}) \times 100$
BR008	Scores must be calculated automatically
BR009	Grade classification must follow predefined grading criteria

5. Performance Classification Rules

Performance Band	Criteria
Top Performer	Percentage ≥ 90
High Performer	75–89
Average	50–74
Needs Support	35–49
At Risk	<35

6. Risk Identification Rules

Rule ID	Business Rule
BR010	Students with low average scores are considered risk candidates
BR011	Poor attendance increases risk level
BR012	Students showing declining performance require intervention

7. Chapter Analytics Rules

Rule ID	Business Rule
BR013	Chapter difficulty is determined using question accuracy
BR014	Chapters below defined accuracy thresholds require review
BR015	Teachers should receive chapter-level insights

8. Dashboard Rules

Rule ID	Business Rule
BR016	Management dashboard should show overall academic performance
BR017	Teacher dashboard should show subject-specific insights
BR018	Student information must follow privacy access rules

9. Data Quality Rules

Rule ID	Business Rule
BR019	Student IDs must not contain duplicates
BR020	Missing assessment records must be flagged
BR021	Invalid scores must be rejected

10. Conclusion

The Business Analysis phase identifies the current operational challenges, root causes, and improvement opportunities. These documents provide the foundation for defining requirements and designing the future Student Performance Analytics & Decision Support System.